

# The Borel dichromatic number

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# Outline

1. **Background:** Borel chromatic numbers and basis problems
2. **Dichromatic number:** a directed analogue
3. **Classical reduction:** from hypergraphs to digraphs
4. **Borel lifting:** Thornton's coding framework
5. **Open questions**

# Borel graphs

A **Borel graph**  $G = (X, E)$  consists of:

- a standard Borel space  $X$  (vertex set),
- a Borel symmetric irreflexive relation  $E \subseteq X^2$  (edges).

A **Borel  $k$ -coloring** is a Borel map  $c: X \rightarrow k$  with  $c(x) \neq c(y)$  for all  $(x, y) \in E$ .

The **Borel chromatic number**  $\chi_B(G)$  is the minimum  $k$  admitting a Borel  $k$ -coloring.

Basic fact for finite graphs; proved in KST '99 in the Borel setting

Every graph of degree  $\leq d$  admits a  $(d + 1)$ -coloring.

# Why Borel?

Every Borel graph has a (possibly non-Borel) proper coloring.

The point: **requiring the coloring to be Borel** makes the problem genuinely harder.

## Example 1

There exist Borel graphs with  $\chi(G) = 2$  but  $\chi_B(G) > \aleph_0$ .

The Borel chromatic number captures a notion of *definable complexity* of the coloring problem.

# The $\mathbb{G}_0$ -dichotomy (Kechris–Solecki–Todorćević, 1999)

## Theorem 2 (KST '99)

*There exists a Borel graph  $\mathbb{G}_0$  such that, for every analytic graph  $G$ :*

$$\chi_B(G) > \aleph_0 \iff \mathbb{G}_0 \leq_B G$$

*where  $\leq_B$  denotes the existence of a Borel homomorphism.*

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*where  $\leq_B$  denotes the existence of a Borel homomorphism.*

In other words:  $\{\mathbb{G}_0\}$  is a **one-element basis** for the class of Borel graphs with uncountable Borel chromatic number.

**Interpretation:** To prove  $\chi_B(G) > \aleph_0$ , there is essentially *one* obstruction to find. The problem is “well-structured.”

# What is a basis?

Fix a class  $\mathcal{C}$  of Borel graphs (e.g. those with  $\chi_B \geq k$ ).

A family  $\mathbf{B}$  is a **basis** for  $\mathcal{C}$  under Borel homomorphism if:

$$G \in \mathcal{C} \iff \exists H \in \mathbf{B}, H \leq_B G.$$

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**Small basis = “simple” decision problem:**

- Size 1: one obstruction to check (like  $\mathbb{G}_0$ )
- Finite: finitely many obstructions
- Countable: “tame” complexity
- Uncountable or none: the problem is genuinely complex



# The $\mathbb{L}_0$ -dichotomy

## Theorem 3 (Carroy–Miller–Schrittesser–Vidnyánszky)

*There exists a Borel graph  $\mathbb{L}_0$  such that, for every analytic graph  $G$ :*

$$\chi_B(G) > 2 \iff \mathbb{L}_0 \leq_B G.$$

Again a one-element basis, this time for the 2 vs. 3-color threshold.

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**So far:** at both the 2/3-color threshold and the countable/uncountable threshold, there is a dichotomy with a one-element basis.

**Question:** Does this continue for higher thresholds?

# No dichotomy at higher thresholds

## Theorem 4 (Todorčević–Vidnyánszky, 2021)

*For every  $k \geq 3$ , the set of (codes for) Borel graphs admitting a Borel  $k$ -coloring is  $\Sigma_2^1$ -complete.*

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## Corollary 5

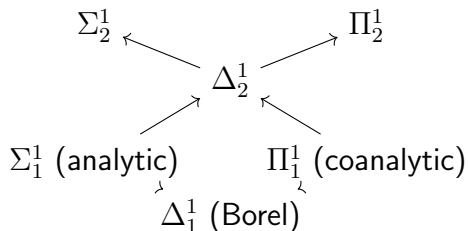
*For  $k \geq 3$ : no countable basis for the class of Borel graphs with  $\chi_B(G) > k$ .*

The landscape for *undirected* Borel chromatic numbers is completely understood:

|                        |                    |
|------------------------|--------------------|
| $\chi_B > 2$           | basis of size 1    |
| $\chi_B > k, k \geq 3$ | no countable basis |
| $\chi_B > \aleph_0$    | basis of size 1    |

# Interlude: what does $\Sigma_2^1$ -completeness mean?

Recall the **projective hierarchy**:



A set  $A \subseteq \omega$  is  $\Sigma_2^1$  if there is a  $\Pi_1^1$  set  $P \subseteq \omega \times \mathcal{N}$  with:

$$n \in A \iff \exists x \in \mathcal{N}, (n, x) \in P.$$

## $\Sigma_2^1$ -completeness: the key example

Why “admits a Borel  $k$ -coloring” is  $\Sigma_2^1$

Given a *nice code*  $e_G$  for a Borel graph  $G$ :

$$\text{“}G\text{ admits a Borel }k\text{-coloring”} \iff \exists e_c \in \mathcal{N} \underbrace{\left[ \text{“}\mathcal{D}_{e_c} \text{ is a } k\text{-coloring of } G\text{”} \right]}_{\Pi_1^1 \text{ in } e_c, e_G}$$

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The inner condition is  $\Pi_1^1$  because it asserts:

- totality:  $\forall x \in V(G), \exists! i < k, c(x) = i$   $[\Pi_1^1]$
- properness:  $\forall (x, y) \in E(G), c(x) \neq c(y)$   $[\Pi_1^1]$

The outer  $\exists e_c$  quantifier makes the whole thing  $\Sigma_2^1$ .

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$\Sigma_2^1$ -complete means: this set is *as hard as any*  $\Sigma_2^1$  set. Every  $\Sigma_2^1$  set reduces to it via a  $\Delta_1^1$  function.



# Why does $\Sigma_2^1$ -completeness destroy countable bases?

## Proof sketch (for chromatic number).

Suppose  $\mathbf{B} = \{H_1, H_2, \dots\}$  is a countable basis for  $\chi_B(G) > k$ . Then:

$$\chi_B(G) > k \iff \exists n, \exists \text{ Borel homomorphism } H_n \rightarrow G.$$

For each fixed  $H_n$ , “ $\exists$  Borel hom  $H_n \rightarrow G$ ” is  $\Sigma_2^1$  in the code for  $G$ .

A countable union of  $\Sigma_2^1$  sets is  $\Sigma_2^1$ .

But the left-hand side  $\{\text{code}(G) : \chi_B(G) > k\}$  is  $\Pi_2^1$ -complete.

This gives  $\Pi_2^1 \subseteq \Sigma_2^1$ , contradicting the strictness of the projective hierarchy.  $\square$

# The dichromatic number (Neumann-Lara, 1982)

Let  $D = (V, A)$  be a directed graph.

A **dicoloring** is a map  $c: V \rightarrow k$  such that every color class induces an *acyclic* subdigraph (no directed cycles).

The **dichromatic number**  $\vec{\chi}(D)$  is the minimum such  $k$ .

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## Example 6

- A tournament on 3 vertices forming a directed 3-cycle:  $\vec{\chi} = 2$ .
- A directed graph with no directed cycles:  $\vec{\chi} = 1$ .
- If  $D$  is a symmetric digraph ( $\{u, v\} \in E \Leftrightarrow (u, v), (v, u) \in A$ ):  
 $\vec{\chi}(D) = \chi(\text{underlying graph})$ .

# Dichromatic vs. chromatic: key difference

## Classical complexity

- “Is  $\chi(G) \leq 2$ ?” is decidable in polynomial time (bipartiteness).
- “Is  $\vec{\chi}(D) \leq 2$ ?” is **NP-complete** (Bokal et al., 2004).

The dichromatic number is “harder” than the chromatic number: complexity starts already at the 2/3-color threshold.

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**Key observation:** Acyclicity is *not* a local constraint.

Checking that a color class is independent (for chromatic number) requires looking at edges one at a time.

Checking that a color class is acyclic may require following long directed paths.

In particular, dichromatic coloring is **not a CSP** since acyclicity cannot be expressed as a homomorphism to a fixed finite template.

# Borel dichromatic number

For Borel directed graphs, define  $\vec{\chi}_B(D)$  analogously, requiring the dicoloring to be Borel.

## Theorem 7 (Raghavan–Xiao, 2024)

*There exists a family  $\{H_\alpha\}_{\alpha < \mathfrak{c}}$  of Borel digraphs, of size continuum, such that:*

$$\vec{\chi}_B(D) > \aleph_0 \iff \exists \alpha, H_\alpha \leq_B D.$$

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This is the directed analogue of the  $\mathbb{G}_0$ -dichotomy, but the basis has *continuum* many elements instead of one.

**Open:** Is  $\mathfrak{c}$  optimal, or could a smaller (perhaps countable) basis exist?

# The main result

## Proposition 8

*The set of nice codes for Borel directed graphs admitting a Borel 2-dicoloring is  $\Sigma_2^1$ -complete.*

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## Recovering the chromatic number

Given an undirected graph  $G$ , symmetrize it:  $\vec{G}$  has arcs  $(u, v)$  and  $(v, u)$  for each edge  $\{u, v\}$ . Then  $\vec{\chi}_B(\vec{G}) = \chi_B(G)$ .

This reduction is  $\Delta_1^1$  on codes, so  $\Sigma_2^1$ -completeness of  $k$ -dicoloring for  $k \geq 4$  follows from Todorčević–Vidnyánszky.

# The complete picture

| Class                           | Basis?                         | Reference        |
|---------------------------------|--------------------------------|------------------|
| $\chi_B(G) > \aleph_0$          | Yes, size 1 ( $\mathbb{G}_0$ ) | KST '99          |
| $\chi_B(G) > 2$                 | Yes, size 1 ( $\mathbb{L}_0$ ) | CMSV             |
| $\chi_B(G) > k, k \geq 3$       | No ( $\Pi_2^1$ -complete)      | TV '21           |
| $\vec{\chi}_B(D) > \aleph_0$    | Yes, size $\mathfrak{c}$       | RX '24           |
| $\vec{\chi}_B(D) > k, k \geq 2$ | No ( $\Pi_2^1$ -complete)      | <b>this talk</b> |

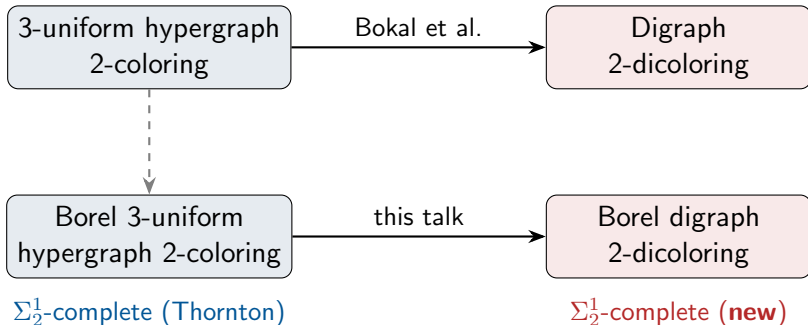
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The case  $k = 2$  for directed graphs is the **last missing piece**.

The proof: lift the classical NP-completeness reduction to the Borel setting.

# Strategy



# Step 1: The classical starting point

## Fact 10

*The problem of deciding whether a 3-uniform hypergraph admits a 2-coloring is NP-complete.*

A **3-uniform hypergraph**  $H = (V, E)$  has hyperedges  $e \in \binom{V}{3}$ .

A **2-coloring** is a map  $c: V \rightarrow 2$  such that no hyperedge is monochromatic.

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**Goal:** Reduce this to 2-dicoloring of directed graphs.

**Idea:** Build a directed graph  $D(H)$  from  $H$  using a finite *gadget* that translates “no monochromatic hyperedge” into “no monochromatic directed cycle.”

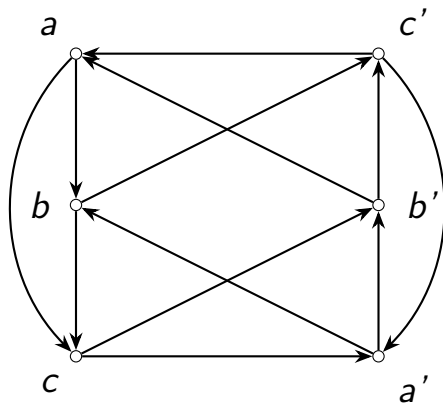
# The template gadget

The gadget  $F$  has:

- **Shared vertices**  $\{a, b, c\}$  (will become hypergraph vertices)
- **Auxiliary vertices**  $\{a', b', c'\}$  (local to each gadget copy)

## Lemma 11 (Gadget property)

A map  $\sigma: \{a, b, c\} \rightarrow 2$  extends to a 2-dicoloring of  $F$  iff  $\sigma$  is **not constant**.





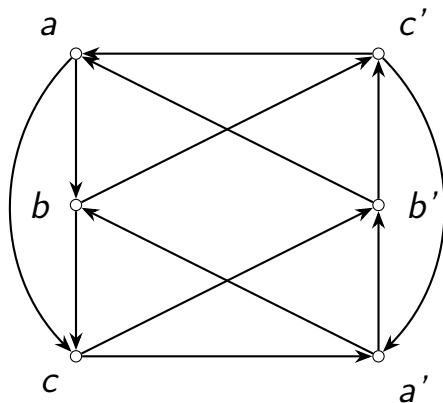
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| coloring of $\{a, b, c\}$                | extends to $F$ ? |
|------------------------------------------|------------------|
| $(0, 0, 1), (0, 1, 0), (1, 0, 0), \dots$ | Yes              |
| $(0, 0, 0)$ or $(1, 1, 1)$               | No               |

# The reduction $H \mapsto D(H)$

Fix a 3-uniform hypergraph  $H = (V, E)$  and a linear order on  $V$ .

**Sorted hyperedges:**  $E_{\text{sort}} = \{(v_1, v_2, v_3) \in E : v_1 < v_2 < v_3\}$ .

**Vertex set of  $D(H)$ :**

$$V_D = V \cup \{(e, t) : e \in E_{\text{sort}}, t \in \{a', b', c'\}\}$$

**Arc set:** For each  $e = (v_1, v_2, v_3)$ , place a copy  $F_e$  of the gadget:

- shared labels  $a, b, c \mapsto v_1, v_2, v_3$
- auxiliary labels  $a', b', c' \mapsto (e, a'), (e, b'), (e, c')$

**Key:** Shared vertices are identified across gadgets. Auxiliary vertices are disjoint.

## Forward direction: $H$ 2-colorable $\Rightarrow D(H)$ 2-dicolorable

Let  $c: V \rightarrow 2$  be a 2-coloring of  $H$ .

For each hyperedge  $e$ : the shared vertices get a non-constant coloring, so by the gadget property it **extends** to a 2-dicoloring of  $F_e$ .

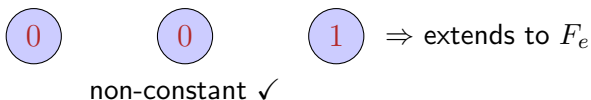
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## Reverse direction: $D(H)$ 2-dicolorable $\Rightarrow H$ 2-colorable

Let  $\psi: V_D \rightarrow 2$  be a 2-dicoloring of  $D(H)$ .

Define  $\varphi = \psi|_V$  (just **restrict** to the original vertices).

For each hyperedge  $e = (v_1, v_2, v_3)$ :

- $F_e$  is a subdigraph of  $D(H)$
- So  $\psi|_{V(F_e)}$  is a valid 2-dicoloring of  $F_e$
- By the gadget property:  $\varphi(v_1), \varphi(v_2), \varphi(v_3)$  are **not all equal**

Hence  $\varphi$  is a 2-coloring of  $H$ . □

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### Important observation

The reverse direction is a *plain restriction*. No choice, no selection, no local finiteness needed. This will matter in the Borel setting.

# Borel lifting: the goal

## Fact 12 (Thornton)

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### Why “ $\Delta_1^1$ on codes”?

A  $\Delta_1^1$  reduction from a  $\Sigma_2^1$ -complete set preserves  $\Sigma_2^1$ -hardness.

So if the reduction is  $\Delta_1^1$  and the equivalence holds, the target is also  $\Sigma_2^1$ -hard.

# Thornton's coding framework (brief)

Fix a good  $\omega$ -parametrization  $U \subseteq \omega \times \mathcal{N}$  of  $\Pi_1^1$ .

A **nice code** for a  $\Delta_1^1$  set  $B$  is an index  $e$  such that  $B$  is simultaneously described by a  $\Pi_1^1$  formula and a  $\Sigma_1^1$  formula, and these descriptions agree.

**The key tool:**

## Lemma 13 (Toolkit)

*The following operations are  $\Delta_1^1$  in the codes:*

- |                                |                                                    |
|--------------------------------|----------------------------------------------------|
| 1. $(A, B) \mapsto A \cup B$   | 4. $A \mapsto \text{dom}(A)$                       |
| 2. $(A, B) \mapsto A \cap B$   | 5. $(A, f) \mapsto f(A)$ , $f$ <i>ctble-to-one</i> |
| 3. $(A, B) \mapsto A \times B$ | 6. $A \mapsto \text{enum. of finite sections}$     |

**Recipe:** Express the construction using these operations  $\Rightarrow$  it is  $\Delta_1^1$  on codes.

# The construction is $\Delta_1^1$ on codes

Let  $H$  be a locally finite Borel 3-uniform hypergraph, coded by  $(V, E)$ . Fix a  $\Delta_1^1$  linear order  $\preceq$  on  $\mathcal{N}$ .

## Step 1: Sort hyperedges.

$$E_{\text{sort}} = \{(a, b, c) \in E : a \preceq b \preceq c\}$$

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## Step 3: Build arc set. (next slide)

## Building the arc set

Define the **substitution map**  $\varphi_H: E_{\text{sort}} \times E_T \rightarrow V_D^2$ :

$$(e, (u, v)) \longmapsto (\text{corresponding arc of } D(H))$$

- shared label in position  $i \mapsto$  vertex  $v_i$  from  $e = (v_1, v_2, v_3)$  (projection)
- auxiliary label  $\ell \mapsto (e, \ell)$  (pairing)

## Building the arc set

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$$(e, (u, v)) \longmapsto (\text{corresponding arc of } D(H))$$

- shared label in position  $i \mapsto$  vertex  $v_i$  from  $e = (v_1, v_2, v_3)$  (projection)
- auxiliary label  $\ell \mapsto (e, \ell)$  (pairing)

$\varphi_H$  is  $\Delta_1^1$  in the code of  $H$ : it uses projections and products, which are recursive.

$\varphi_H$  is **finite-to-one**:  $|E_T|$  is fixed, and each vertex appears in finitely many hyperedges (local finiteness).

Arc set:  $A_D = \varphi_H(E_{\text{sort}} \times E_T) \Rightarrow$  toolkit item (5) ✓

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### Lemma 14

*The map  $\text{code}(H) \mapsto \text{code}(D(H))$  is  $\Delta_1^1$ .*



## Borel forward direction: Luzin–Novikov

### Claim 15

*If  $H$  is Borel 2-colorable, then  $D(H)$  is Borel 2-dicolorable.*

# Borel forward direction: Luzin–Novikov

## Claim 15

*If  $H$  is Borel 2-colorable, then  $D(H)$  is Borel 2-dicolorable.*

## Proof.

Let  $c: V \rightarrow 2$  be a Borel 2-coloring of  $H$ . For each  $e \in E_{\text{sort}}$ , at least one extension to a 2-dicoloring of  $F_e$  exists.

Define the set of valid local completions:

$$S = \{(e, \sigma) \in E_{\text{sort}} \times \mathcal{C}_F : \sigma \text{ extends to a 2-dicoloring of } F_e\}$$

$S$  is Borel, with **finite** nonempty sections  $S_e$ . By the **Luzin–Novikov theorem**:  
 $\exists$  Borel selector  $e \mapsto \sigma(e) \in S_e$ .

Pasting these selections with  $c$  on shared vertices gives a Borel 2-dicoloring of  $D(H)$ . □

## Borel reverse direction: just restrict

### Claim 16

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# Borel reverse direction: just restrict

## Claim 16

*If  $D(H)$  is Borel 2-dicolorable, then  $H$  is Borel 2-colorable.*

## Proof.

Let  $\psi: V_D \rightarrow 2$  be a Borel 2-dicoloring of  $D(H)$ .

Define  $\varphi = \psi|_V$ .

Since  $V \subseteq V_D$  is Borel,  $\varphi$  is Borel.

For each hyperedge  $e$ :  $F_e \subseteq D(H)$  is a subdigraph, so  $\psi|_{V(F_e)}$  is a valid 2-dicoloring, hence the shared-vertex colors are not all equal.

So  $\varphi$  is a Borel 2-coloring of  $H$ . □

# Proof of the main result

## Proof of Proposition.

**Hardness:** We have a  $\Delta_1^1$  reduction from Borel 2-colorable 3-uniform hypergraphs ( $\Sigma_2^1$ -complete) to Borel 2-dicolorable digraphs. Hence the latter is  $\Sigma_2^1$ -hard.

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**Hardness:** We have a  $\Delta_1^1$  reduction from Borel 2-colorable 3-uniform hypergraphs ( $\Sigma_2^1$ -complete) to Borel 2-dicolorable digraphs. Hence the latter is  $\Sigma_2^1$ -hard.

### Upper bound:

$$\text{"}D \text{ admits a Borel 2-dicoloring"} \iff \exists e_c \underbrace{\left[ \text{"}\mathcal{D}_{e_c} \text{ is a valid 2-dicoloring"} \right]}_{\Pi_1^1}$$

The  $\Pi_1^1$  condition checks:

- totality and range  $\subseteq \{0, 1\}$  ( $\Pi_1^1$ )
- each color class induces no directed cycle ( $\Pi_1^1$ : universal over finite paths)

So the set is  $\Sigma_2^1$ . Combined with hardness:  $\Sigma_2^1$ -complete.  $\square$

# The corollary: no countable basis

## Proof of Corollary.

Suppose  $\mathbf{B} = \{H_1, H_2, \dots\}$  is a countable basis for  $\vec{\chi}_B(D) \geq 3$ :

$$\vec{\chi}_B(D) \geq 3 \iff \exists n, \exists \text{ Borel hom } H_n \rightarrow D.$$

# The corollary: no countable basis

## Proof of Corollary.

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$$\vec{\chi}_B(D) \geq 3 \iff \exists n, \exists \text{ Borel hom } H_n \rightarrow D.$$

For fixed  $H_n$ : “ $\exists \text{ Borel hom } H_n \rightarrow D$ ” is  $\Sigma_2^1$  in the code of  $D$ .

Countable union  $\Rightarrow$  the right-hand side is  $\Sigma_2^1$ .

But the left-hand side is  $\Pi_2^1$ -complete (complement of our  $\Sigma_2^1$ -complete set).

$\Pi_2^1 \subseteq \Sigma_2^1$  contradicts the strictness of the projective hierarchy.  $\square$   $\square$



# A stronger conclusion

The complexity argument rules out more than just a countable basis.

## Observation

If  $\mathbf{B} \in \Sigma_2^1$  as a subset of the code space, then:

- “ $H \in \mathbf{B}$ ” is  $\Sigma_2^1$
- “ $\exists$  Borel hom  $H \rightarrow D$ ” is  $\Sigma_2^1$
- conjunction and  $\exists$ -quantification preserve  $\Sigma_2^1$

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So the right-hand side would be  $\Sigma_2^1$ , contradicting  $\Pi_2^1$ -completeness.

**Conclusion:** Any basis  $\mathbf{B}$  must satisfy  $\mathbf{B} \notin \Sigma_2^1$ . It is at least as complex to describe as the set of all digraphs without a Borel acyclic 2-coloring.

A complexity lower bound destroys all “simple” bases, not just countable ones.

# Open questions I: the uncountable threshold

Raghavan–Xiao give a continuum-size basis for  $\vec{\chi}_B(D) > \aleph_0$ .

**Q1. Is the basis irredundant?** I.e., does removing any single element break the basis? (Likely yes, from pairwise incomparability.)

**Q2. Can the basis be made smaller?** Is  $\mathfrak{c}$  optimal?

**Q3. Is there a direct (non-complexity) proof** that no *countable* basis exists for  $\vec{\chi}_B(D) > \aleph_0$ ? Perhaps via diagonalization, adapting Miller's anti-basis argument for  $G_f$  graphs.

Note: the complexity approach does *not* work here—the continuum-size basis shows the set is not  $\Pi_2^1$ -complete.

## Open questions II: structure and CSPs

- Q4. Structural restrictions.** Does a countable basis exist for digraphs  $G_f$  (generated by a single Borel function) with  $\vec{\chi}_B(G_f) \geq 3$ ?  
Miller showed no countable basis exists for undirected  $G_f$  with  $\chi_B \geq 3$ .
- Q5. Bounded width CSPs.** Thornton showed that width 1 structures are exactly those where solvable Borel instances have Borel solutions.  
Grebík–Vidnyánszky showed the Borel/classical CSP dichotomies diverge.  
*What is the exact complexity dividing line within bounded width?*
- Q6. Measurable arboricity** (Thornton). Arboricity = undirected analogue of dichromatic number.  
*Can  $\mu$ -measurable arboricity differ from classical arboricity by more than 1?*

Thank you.

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